

Application tips for a heat-sink design

- Interfaces for heat transfer should be smooth and flat. Use thermal grease or interface pads wherever possible.
- Semiconductors should be spaced so as to obtain uniform power density.
- Cabinets and racks should be adequately vented at the top and bottom of the enclosure.
- Maximise expose area to increase surface emissivity.
- Use a high conductivity material for designing a heat-sink.
- The material's thickness and interface areas of enclosure should be adequate to handle expected power density.

mented with. Inorganic ceramics, non-metallic and non-corrosive materials are already being used in LEDs. Heat conduction and electric insulation make these ideal for heat-sink applications.

Thermally-conductive plastics, or thermoplastics that are loaded with a conductive material, such as aluminium nitride are naturally non-corrosive, making these ideal for outdoor applications. Thermoplastics are also 20 to 30 per cent less expensive than aluminium as these are simpler to make.

Italian luminaire manufacturer,

Reggiani Lighting, combines copper and extruded aluminium to reap the best of both metals in LEDs. There are a lot of experiments being done with these materials. Made from graphite, hitherm is a lightweight flexible material. Carbon being an excellent conductor of heat, it enables this material to be a very good heat-sink. However, price point of these solutions is relatively high.

Liquid cooling solutions. There might come a time while designing a system when a simple fan or heat-sink is not enough. Ramanan explains, "If the rate of generation of

Parameters in design

A thermal vent in a system cannot be designed willy-nilly. One of the biggest parameters with thermal solution design is integration with system design and cost. If a thermal solution makes your mobile bulky, it would not be much of a solution now, would it? Some of the other design constraints include:

- Induced approach flow velocity
- Cross-sectional geometry of incoming flow
- Amount of required heat dissipation
- Maximum heat-sink temperature
- Size of heat-sink

Keeping these challenges in mind, the available parameters for a designer to modify are:

- Number of fins
- Thickness of fins
- Shape of fins
- Material

heat is very high, where we cannot go for simpler heat-sinks, we can go for liquid cooling." This calls for solutions that can manage a higher rate of heat transfer.



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Silicone
Synthetic Rubber
Thinners
Strippers

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Silicone
UV Curable

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Electrically Conductive
Thermally Conductive
Epoxy
UV Curable

GASKETING

FIP Conductive
FIP Non Conductive
Preform

THERMAL MANAGEMENT

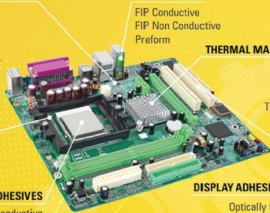
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